

Energy Market

Edge-Based Forecasting and Lifeline Power Buying System

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01 Background

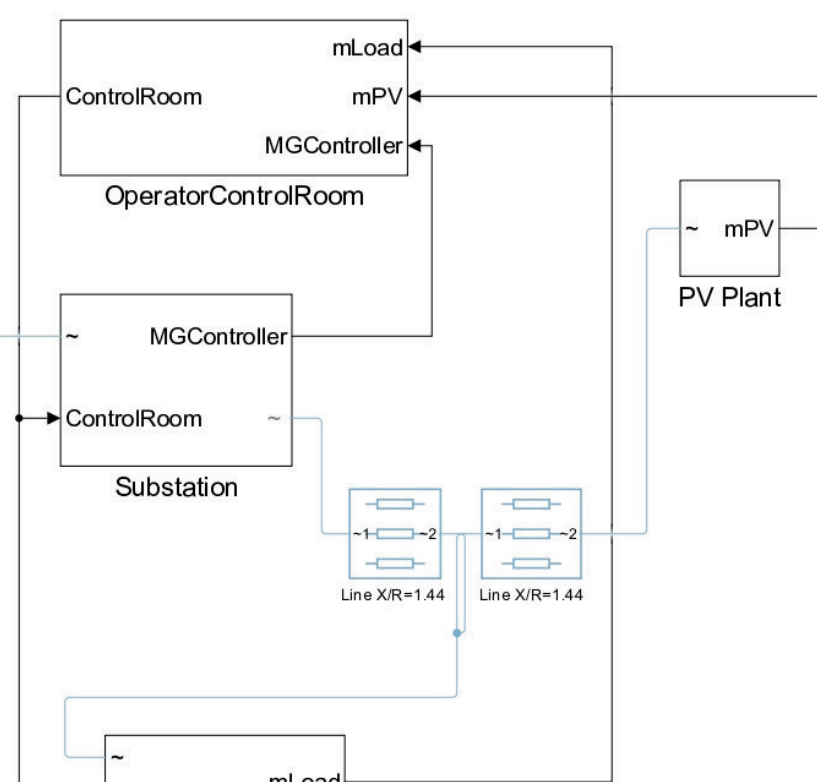
- National Energy Target: Sri Lanka has set an ambitious goal of generating 70% of its electricity from renewable sources (solar and wind) by 2030.
- Renewable energy sources are intermittent and unpredictable, leading to voltage fluctuations and supply-demand mismatches, which threaten grid reliability.
- Currently, distribution licensees are piloting BESS integration into high and low voltage grids to enhance stability and provide ancillary services like frequency and voltage regulation.

03 Solution

- The Lifeline Power Buying System is a software solution designed to connect battery energy storage owners with consumers who need essential power during outages. This platform enables reliable access to power during outages and provides BESS owners with monetization opportunities.
- A MATLAB simulation models grid behavior to evaluate platform performance and validate control strategies.
- An energy consumption forecasting system with real-time edge implementation enhances grid monitoring and stability by providing low-latency support critical to grid modernization.

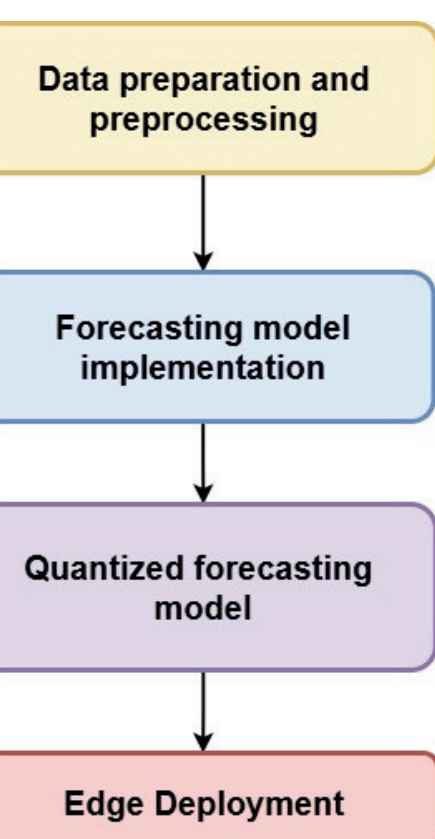
05 Methodology

Grid Simulation



- The simulation is designed and tested using MATLAB Simulink, a professional platform commonly used for power grid simulations.
- Key components – the main grid, substation, PV plant, BESS, consumer loads, and controllers.
- The system controller is developed in Python. It uses the TCP protocol to communicate with the power grid and REST APIs to interact with the lifeline platform

Forecasting Model



- Mean squared error is used as the loss function.
 - Mean absolute percentage error is used as the main performance metric.
- $$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{actual_i - predicted_i}{actual_i} \right| \times 100$$

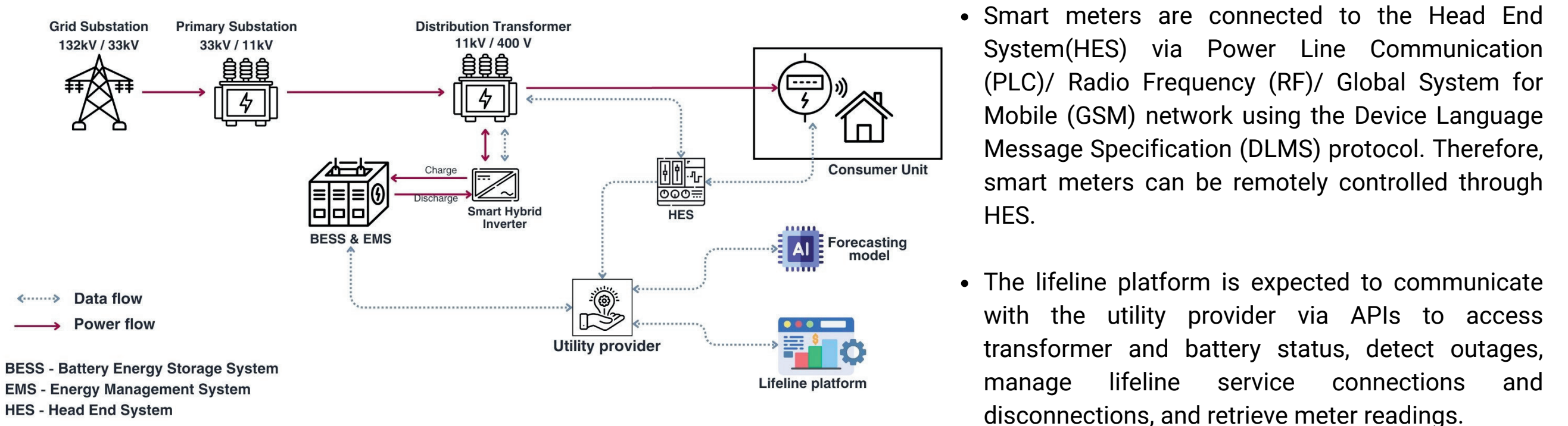
- Energy consumption data is sequential.
- A transformer encoder is used to model the sequential data.

02 Motivation & Problem Statement

- Sudden power cuts can cause significant inconvenience to the public, especially for critical loads, yet there are no standard measures in place to provide power during outages.
- BESS is an underutilized resource during power outages, and effective management of distributed energy resources can lead to grid modernization
- Modern grid control needs accurate local forecasting and monitoring to enable load balancing and optimal BESS use.

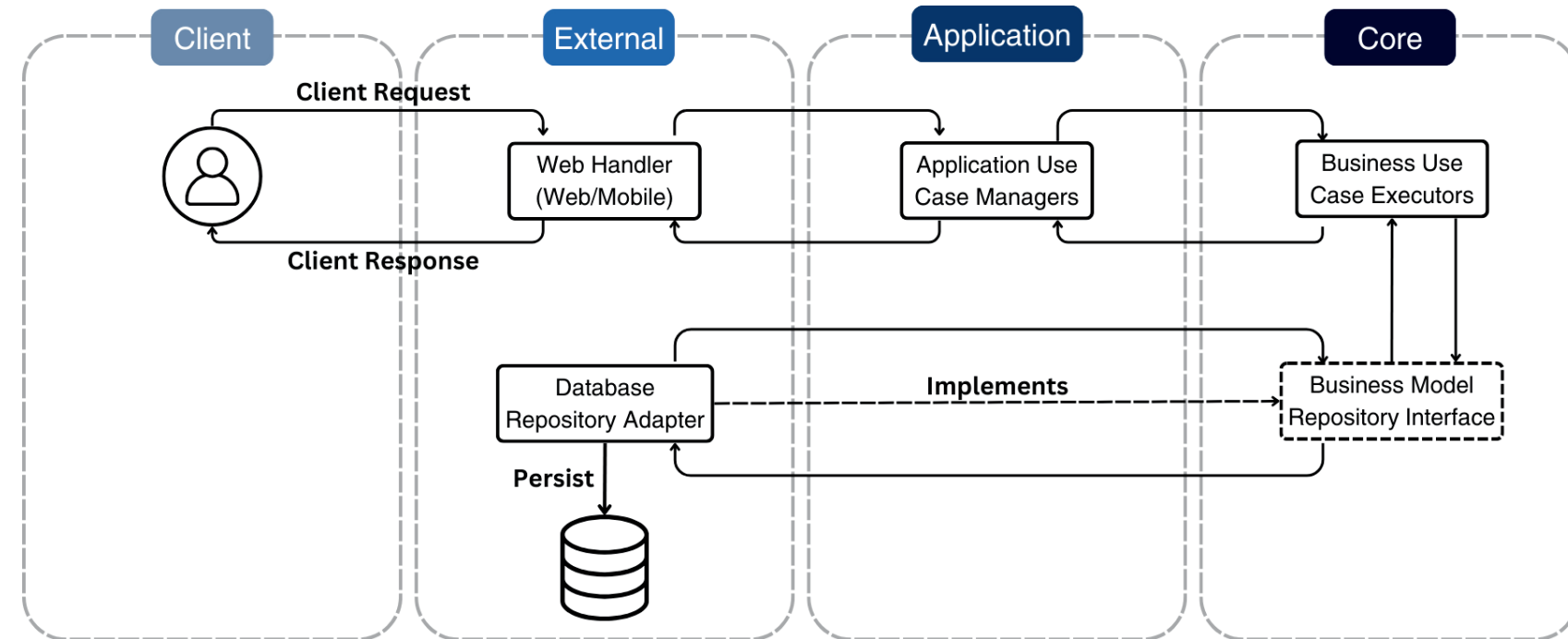
04 Architecture

Overview



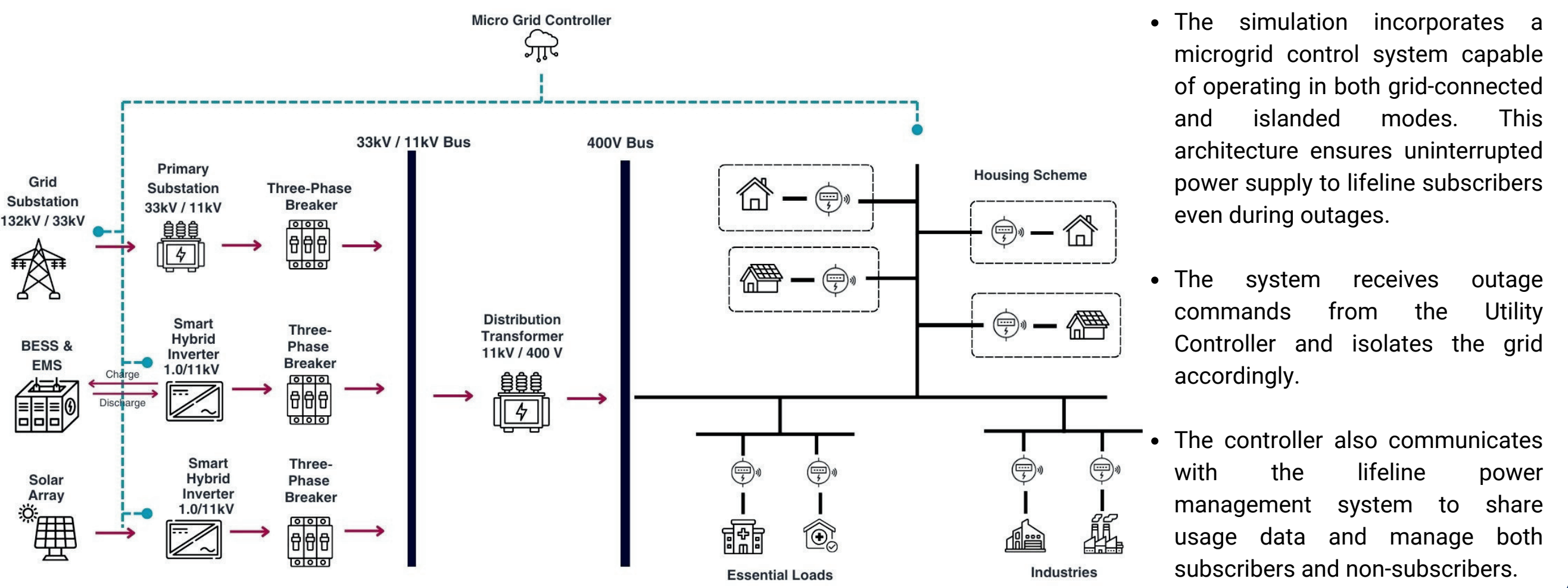
- Smart meters are connected to the Head End System(HES) via Power Line Communication (PLC)/ Radio Frequency (RF)/ Global System for Mobile (GSM) network using the Device Language Message Specification (DLMS) protocol. Therefore, smart meters can be remotely controlled through HES.
- The lifeline platform is expected to communicate with the utility provider via APIs to access transformer and battery status, detect outages, manage lifeline service connections and disconnections, and retrieve meter readings.

Software Architecture



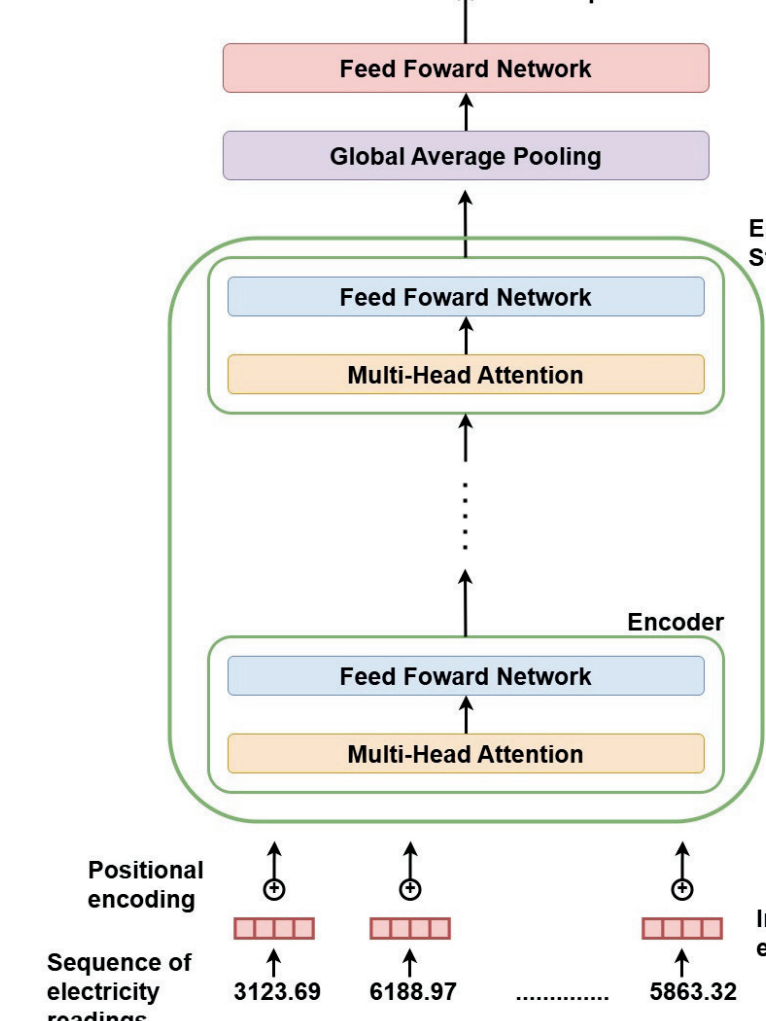
- The system is designed using a Clean Code Architecture approach to ensure clear separation of concerns across layers.
- This structure improves scalability, testability, and maintainability, allowing core business logic to remain independent of external interfaces and infrastructure.
- It supports modular development and enables seamless integration of new technologies without affecting core functionality.

Simulation Architecture



- The simulation incorporates a microgrid control system capable of operating in both grid-connected and islanded modes. This architecture ensures uninterrupted power supply to lifeline subscribers even during outages.
- The system receives outage commands from the Utility Controller and isolates the grid accordingly.
- The controller also communicates with the lifeline power management system to share usage data and manage both subscribers and non-subscribers.

Model Architecture



- The forecasting model is trained to predict the hourly/daily energy consumption.
- This comprises of 6 encoder blocks, each comprising of a self attention layer followed by a feed forward network.
- The self-attention layer uses multi-head attention with 4 attention heads.

Edge Implementation

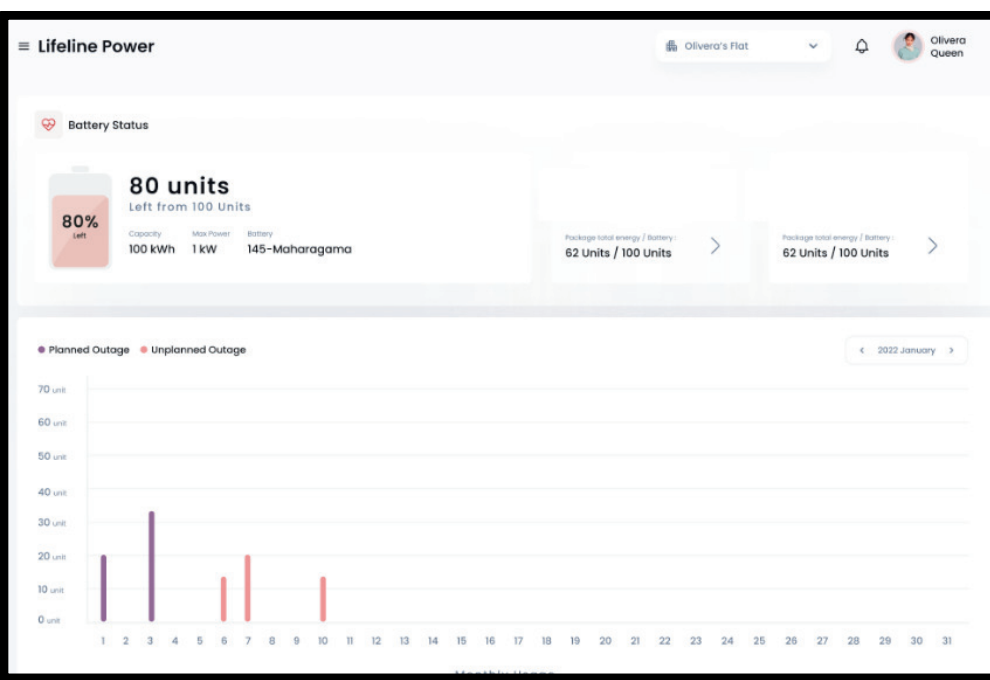
- Post-training quantization approach was employed.
- Quantization techniques used,
 - 16 bit floating point quantization
 - 8 bit integer quantization
- Quantized model was deployed in a Raspberry Pi 4 Model B for edge inference.

06 Technology Stack



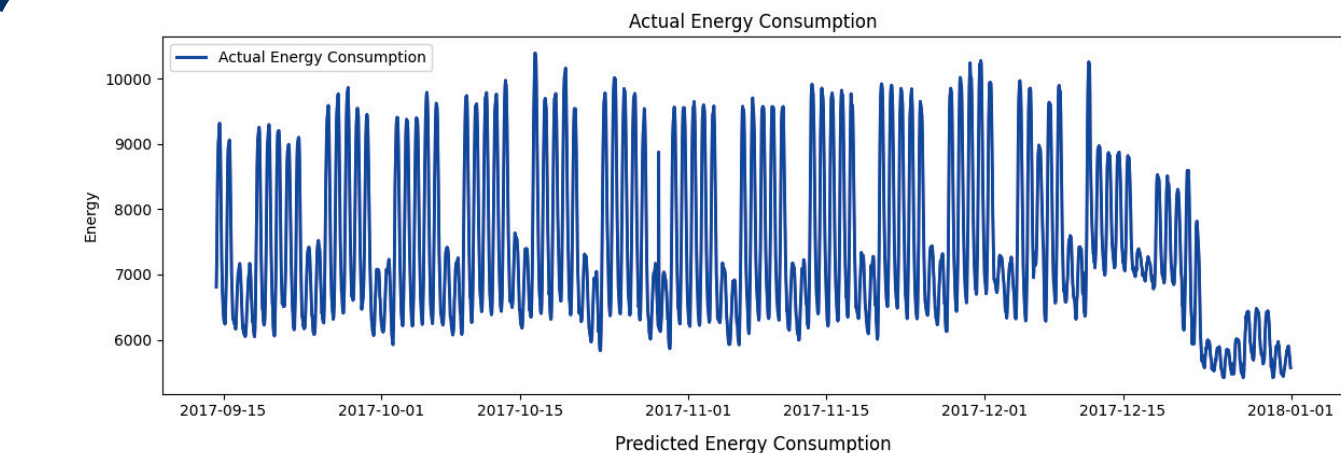
Backend Development	Java
Database	Mongo DB
Cloud Service Provider	Azure
UI Design	Figma
UI Development	React
Simulation and Backend Integration	Python
Source Code Management	Git Hub
CI/CD Pipeline	Git Hub Actions
API Testing	Postman
Simulation	MATLAB

07 Results



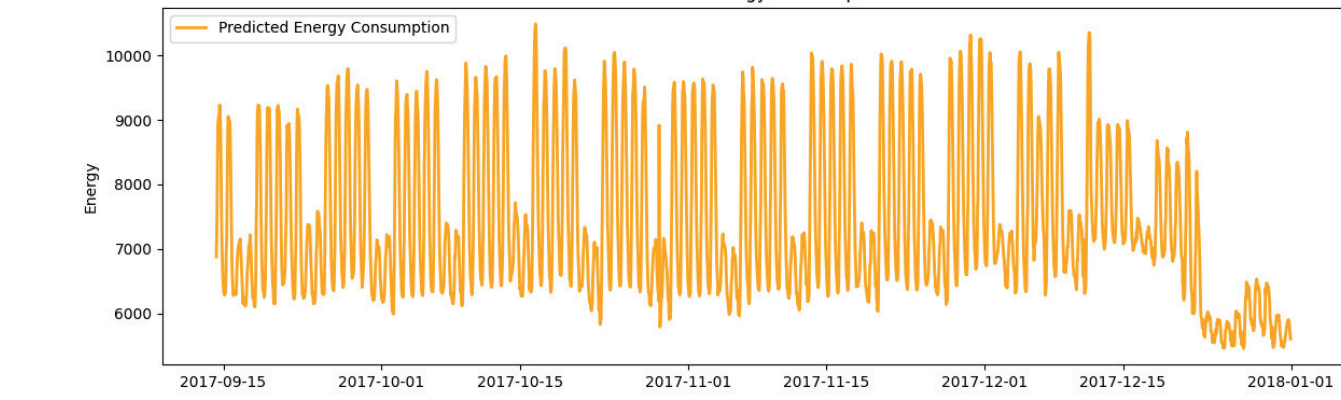
Potential Beneficiaries:

- Consumers with critical loads can ensure a continuous power supply during outages, avoiding losses due to disruptions.
- BESS providers can monetize otherwise unused battery energy during power cuts.

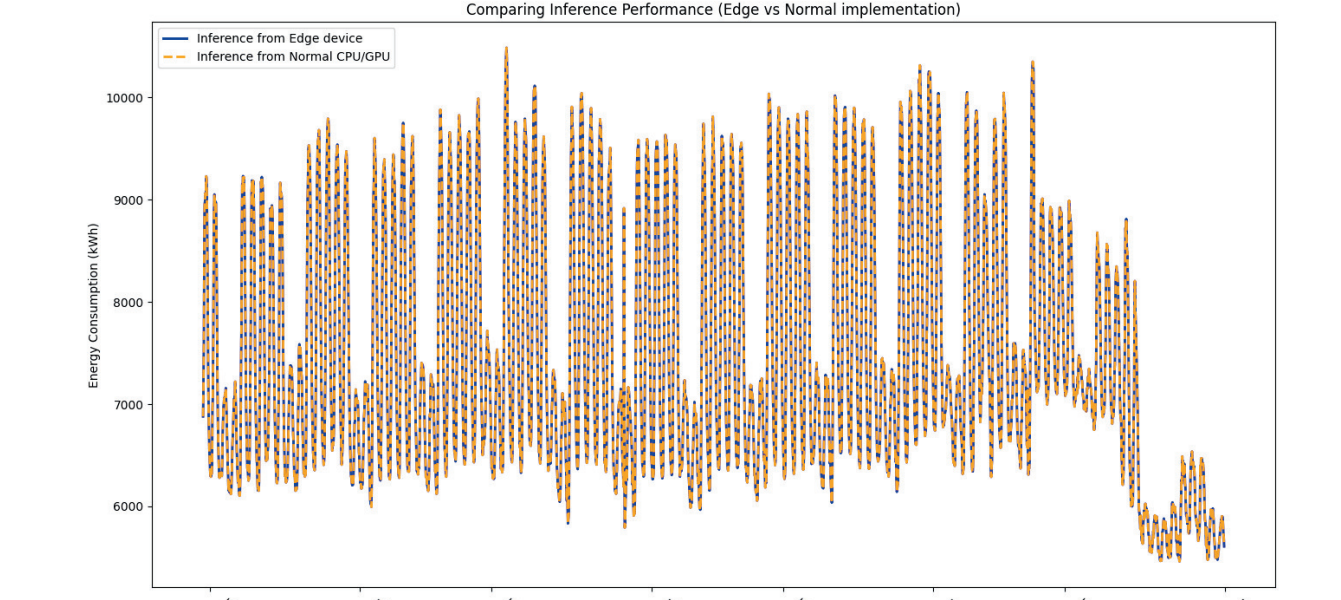


Granularity	Num Features	Time Steps	Test MAPE
Hourly	1	24	1.18
	1	12	1.392
	1	6	24.99
	2	24	1.279
Daily	2	12	1.515
	1	7	2.784
	1	4	4.367
	2	7	3.03
	2	4	4.18

Variation of MAPE values for both daily and hourly forecasts with the number of features used and the time steps considered.



Variation of actual energy consumption vs the predicted energy consumption



The inference performance in the edge device was identical to that on a normal CPU/GPU.

Precision	Model Size Reduction	Inference Speed	Inference Accuracy
16-bit FP	50%	Similar	0.5% reduction
8-bit INT	70%	20% increase	2% reduction

The comparison of quantized model performance against the standard 32 bit floating point precision model.

The following plots show the results from the first 5 seconds of the power grid simulation. At 1 second, an outage is triggered, switching the system from grid-connected mode to islanded mode. At 4 seconds, a heating element is added as a load.

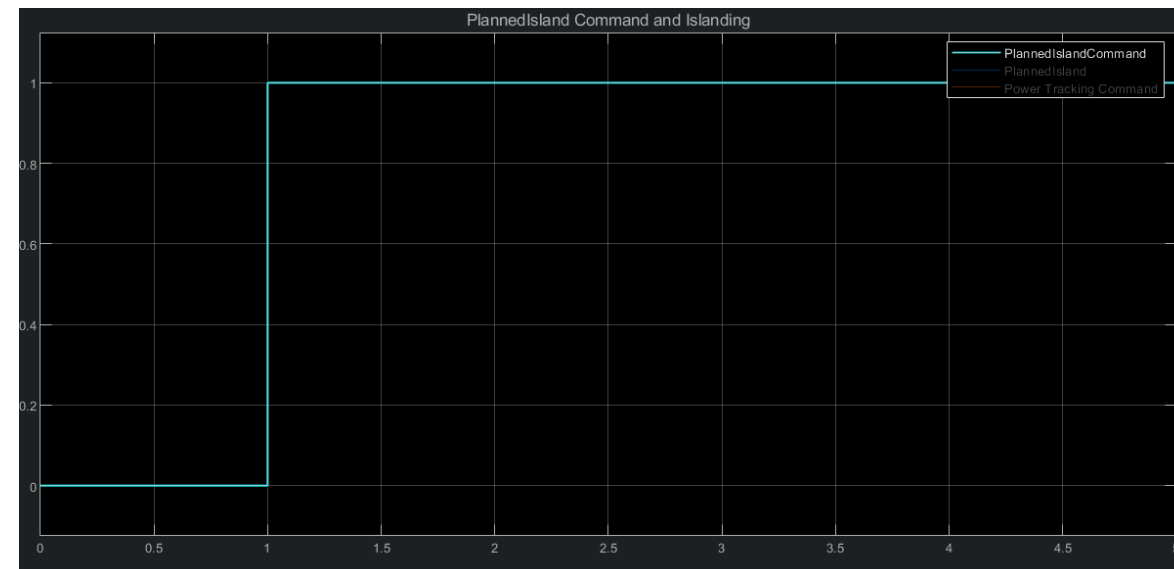


Figure 1.1 Islanding command is triggered at 1 s

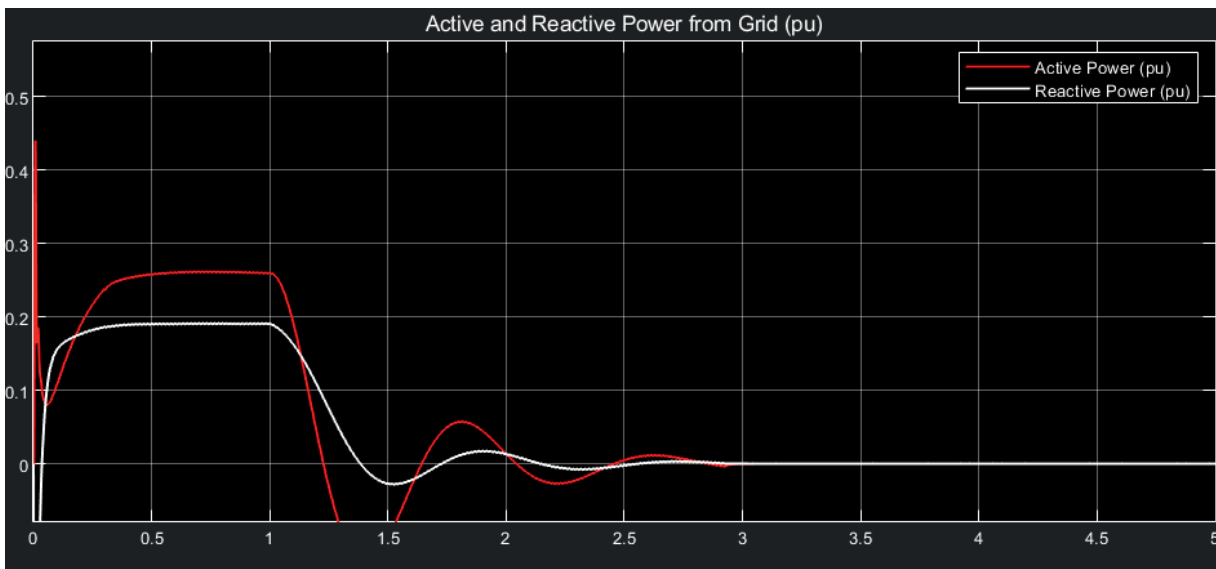


Figure 1.2 Active and reactive power of the main grid drops to zero

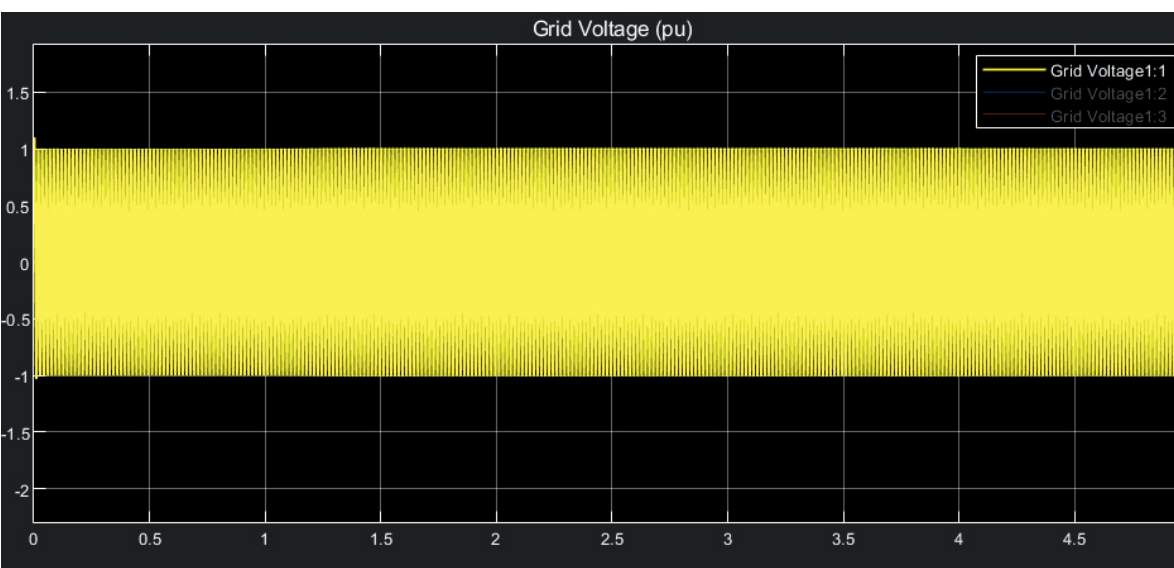


Figure 1.3 Grid voltage remains constant, while grid current drops to zero as the grid is isolated from the main supply.

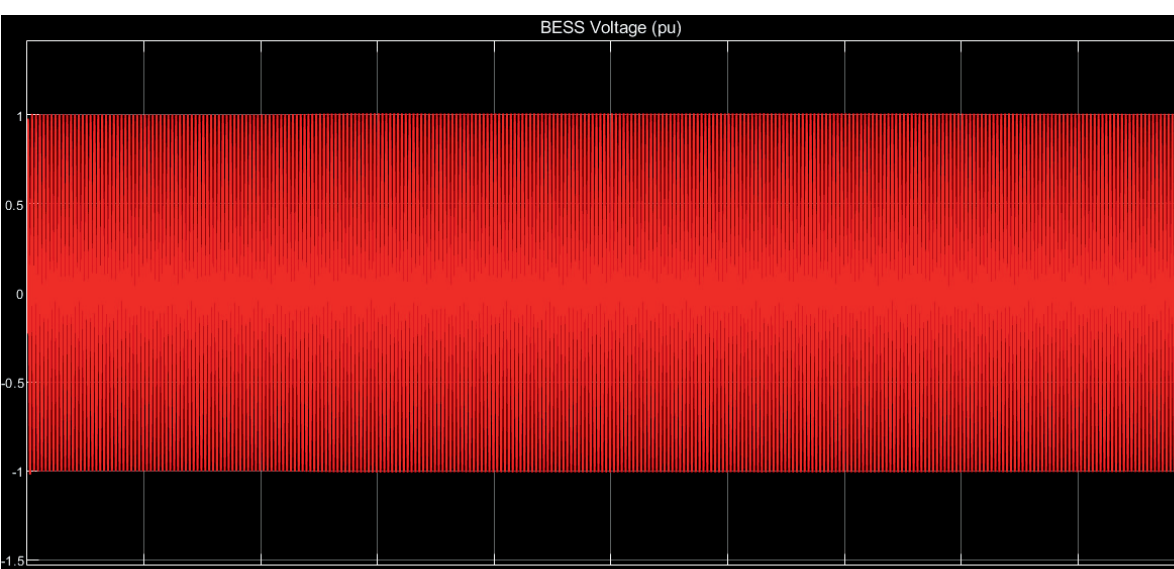
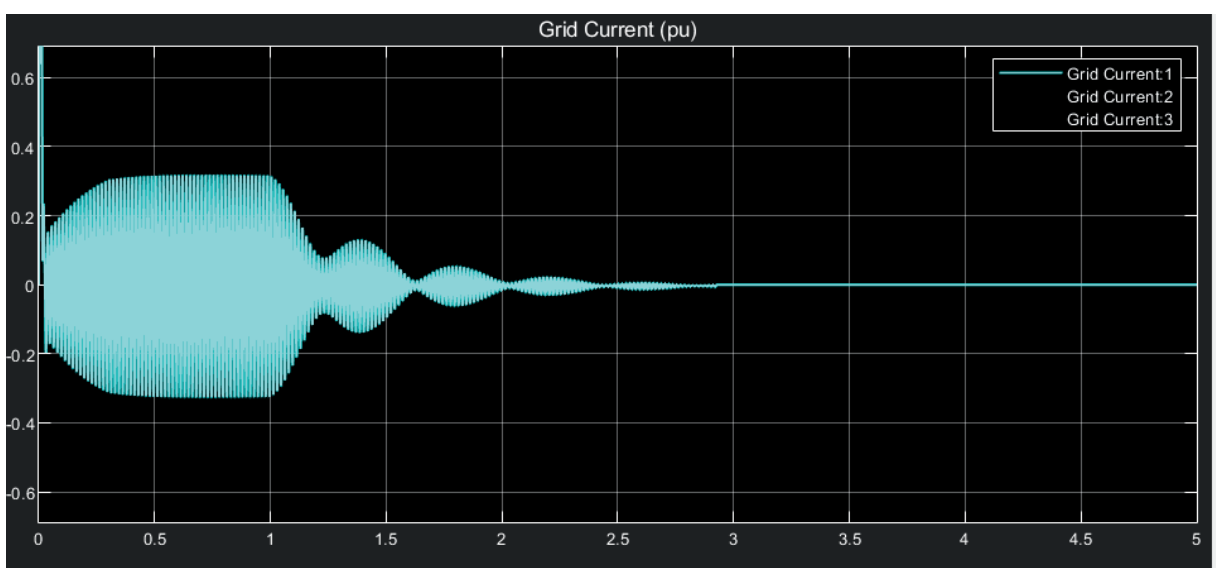


Figure 1.4 BESS voltage remains stable, while BESS current increases to supply the load in islanded mode. The grid current also increases at 4 seconds due to the heating element load.

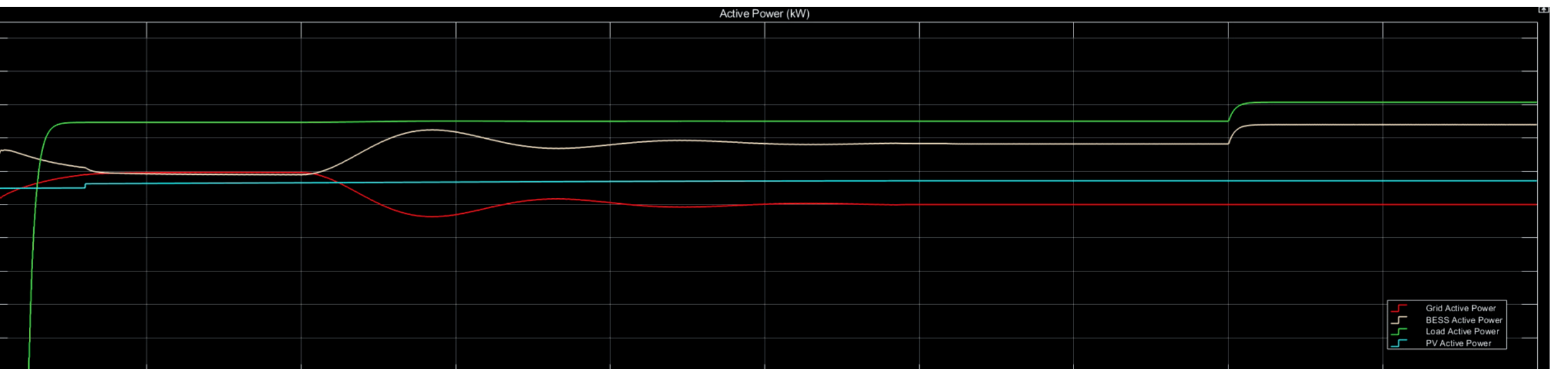


Figure 1.5 Power variation of the grid, BESS, PV plant, and consumer loads: in islanded mode, the grid power drops to zero, while the BESS and PV plant together meet the consumer power demand.

08 Conclusion

- Improved distributed energy resource management, developed edge-based energy forecasting, and implemented a lifeline power-buying platform for outage resilience.
- An energy consumption forecasting model was trained, optimized using quantization, and deployed on a Raspberry Pi 4 for real-time edge-based predictions.
- Addressed gaps in domain knowledge and policy limitations through expert collaboration and plans to validate the system using grid-based simulations.
- The solution demonstrates how underutilized battery storage can enhance grid resilience, potentially influencing future investments and reducing economic losses from power outages.

09 References

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